“I have several hundred coins. Can you classify them automatically?”

A colleague, at my door

Let’s go to Mentimeter



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What I asked

“What do the data look like, and in what form can you give them to me?”

A reasonable question. An engineer’s reflex: find out what you have to work with.

I did not ask **what he needed it for**. Hold that thought for three hours.

His answer

Cleaned

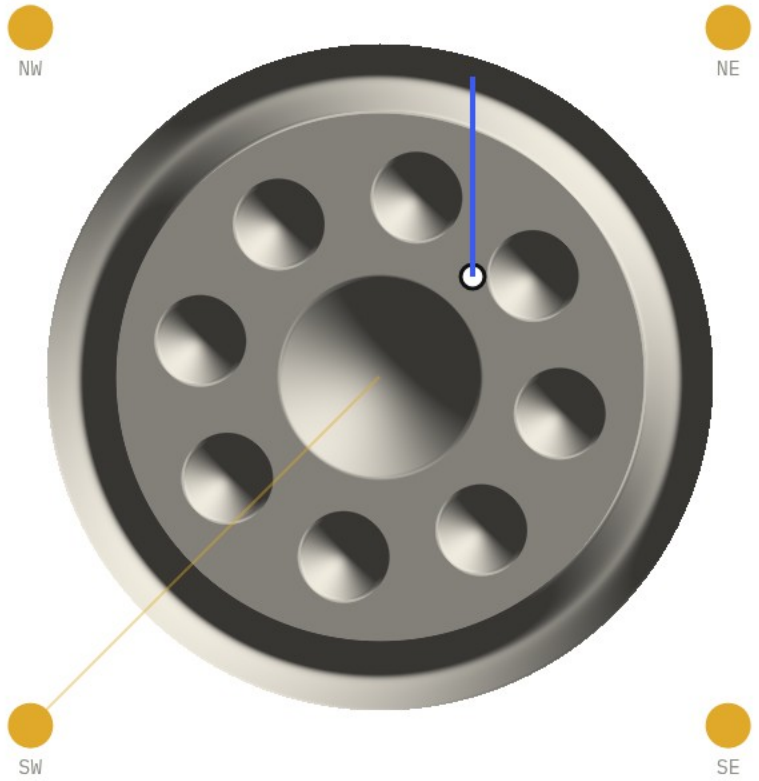
conserved and photographed under controlled light

And in 3D

he could give me the surface of every one of them

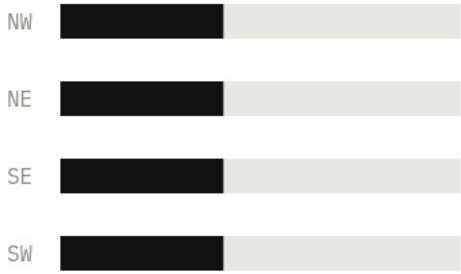
The second one is the part I did not expect. Scanning a coin properly is slow — so how do you scan **hundreds**?

How you scan hundreds of coins

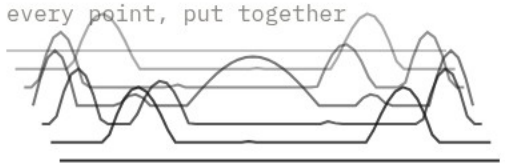


which way that point tilts

brightness of that one point



every point, put together



What we had, before writing a line

Photographs	Surfaces	Several hundred
obverse and reverse, high resolution	the same coins in 3D	one find, one site

Data before the question is a rare luxury. It is also a trap: it makes it easy to do **what the data allow** instead of what the question needs.

ACT II

What kind of material is this

Before choosing a method, look at what you are choosing it for.

I II III IV V VI VII VIII IX X

0:14 — 16 minutes

Several hundred of these



Bořitov. 660 coins, both sides of each. Nothing here has been selected to look good.

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Every one of those is visible to a machine too

A machine sees all of it — and has no way of knowing which parts are the coin and which parts are what happened to it.

Corrosion is structure. A crack is structure. A hard shadow is structure. Nothing in a photograph marks them as **not the point**.

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One layer matters. The rest is noise.



- ✓ **The struck image**
  letters, symbols, the portrait — this is the coin
- × The colour of the metal
  alloy, patina, how it was cleaned
- × Damage
  corrosion, cracks, the edge of the flan
- × The photograph itself
  light direction, shadow, background
- × How it happens to be turned
  nothing to do with the die

What he actually wanted

Sorting several hundred coins by hand is weeks of work — and one person does it, one way.

He did not ask for a model. He asked for the weeks back.

What we know	The goal, the material, and the one feature that decides.
What follows from it	Everything except the struck image is interference.
What it looks like	Like all that is left is picking an algorithm. It is not.

ACT III

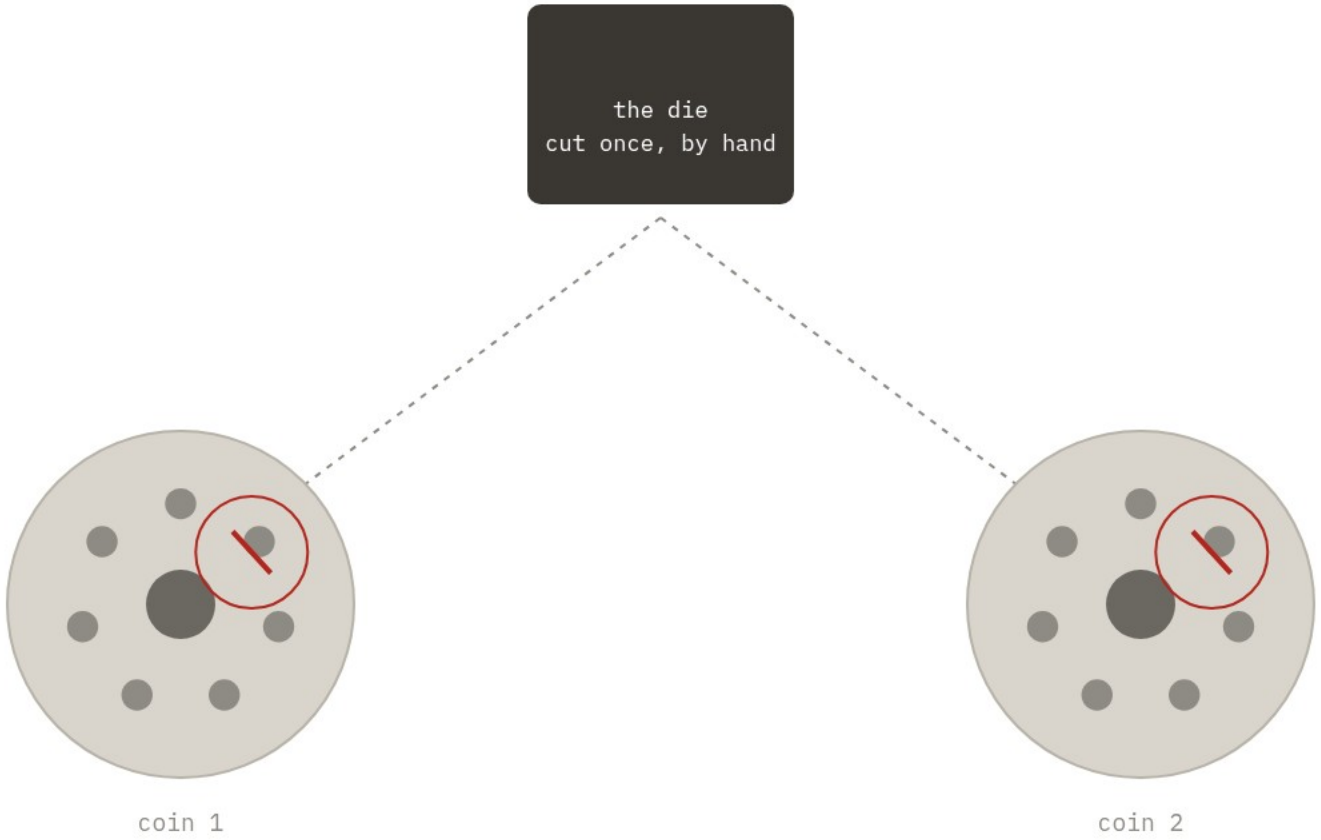
And then he asked something else

Eight minutes. The most expensive eight minutes in the story.

I II **III** IV V VI VII VIII IX X

0:30 — 8 minutes

What a die is



the same flaw in the iron — on both

Why anyone cares

Two coins from one die were in one place, at one moment, in one pair of hands.

Found two hundred kilometres apart, that is a statement about **movement** — and it needs no typology to make it. A classification is an opinion. A shared die is a fact.

“And can you tell me which coins were struck by the same die?”

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I asked nothing.
I said: yeah, why not.

Nobody knew how many dies were in that box — or whether **any two coins shared one at all**. I accepted a task that might have had no answer.

You just asked better questions than I did.

What changed	From “which group is this in” to “did one tool touch both”.
What that is worth	The difference between an opinion and a fact about the past.
What it cost	One sentence, unasked.

ACT IV

The cheapest thing that could work

Not because it is elegant. Because expensive methods have to earn their place.

I II III **IV** V VI VII VIII IX X

0:38 — 24 minutes

Let’s go to Mentimeter



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Three constraints did the choosing

1. **Not an algorithm for this box**
   it has to run on any photograph or scan of any coin in any state — tuning to Bořitov is work thrown away with the next find

2. **Nothing to train on**
   no annotations, not much data, no capacity

3. **One week**
   that is the whole budget

How do you make something that works, out of nothing?

No labels. No training set. No month to spare. This is the ordinary situation, not the hard case.

What we can afford, before what works

coins in the box

660

every coin against every other coin

217,470

$n \times (n - 1) / 2$

a method that takes 0.05 ms a pair

11 seconds

a method that takes 1 second a pair

3 days

a method that takes 15 seconds a pair

38 days

COINS 660

Let’s go to Mentimeter



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Twenty ways to pull relief out of a photograph

1–2	bilateral subtract
3–5	difference of Gaussians, three scales
6–8	top-hat, black-hat, both
9–10	radial gradient removed
11–12	CLAHE
13–14	Sobel, and relief shading
15–20	all of it again, on depth
+	binarised variants of the best eight

Every one of these is still in the service, numbered. None of them was enough.

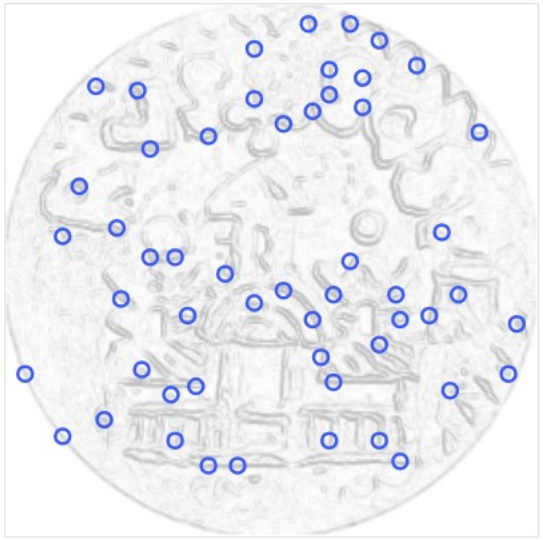
Edges and corners, computed live



the photograph



where brightness changes



53 “interesting points”

Look at where they landed.

On the rim. On the lettering. On the scratches. **Not once on “which emperor is this”.**

Why none of it was enough

They all assume that what matters is **contrast**.

On a worn coin, the strongest contrast is between corrosion and metal — not between the device and the field.

The algorithm found the most striking thing in the image, honestly and correctly. **The most striking thing was the damage.**

What it can do	Find edges, corners and texture, fast, on any image, with no training.
What it is really asking	“Where does brightness change?” — which is the question only on a clean coin.
Where it breaks	The moment damage is more contrasty than the device. Which is most of archaeology.

ACT V

Borrowing an idea from image search

Nobody ever wrote down what
“similar” means. And it works anyway.

I II III IV **V** VI VII VIII IX X

1:14 — 26 minutes

Let’s go to Mentimeter



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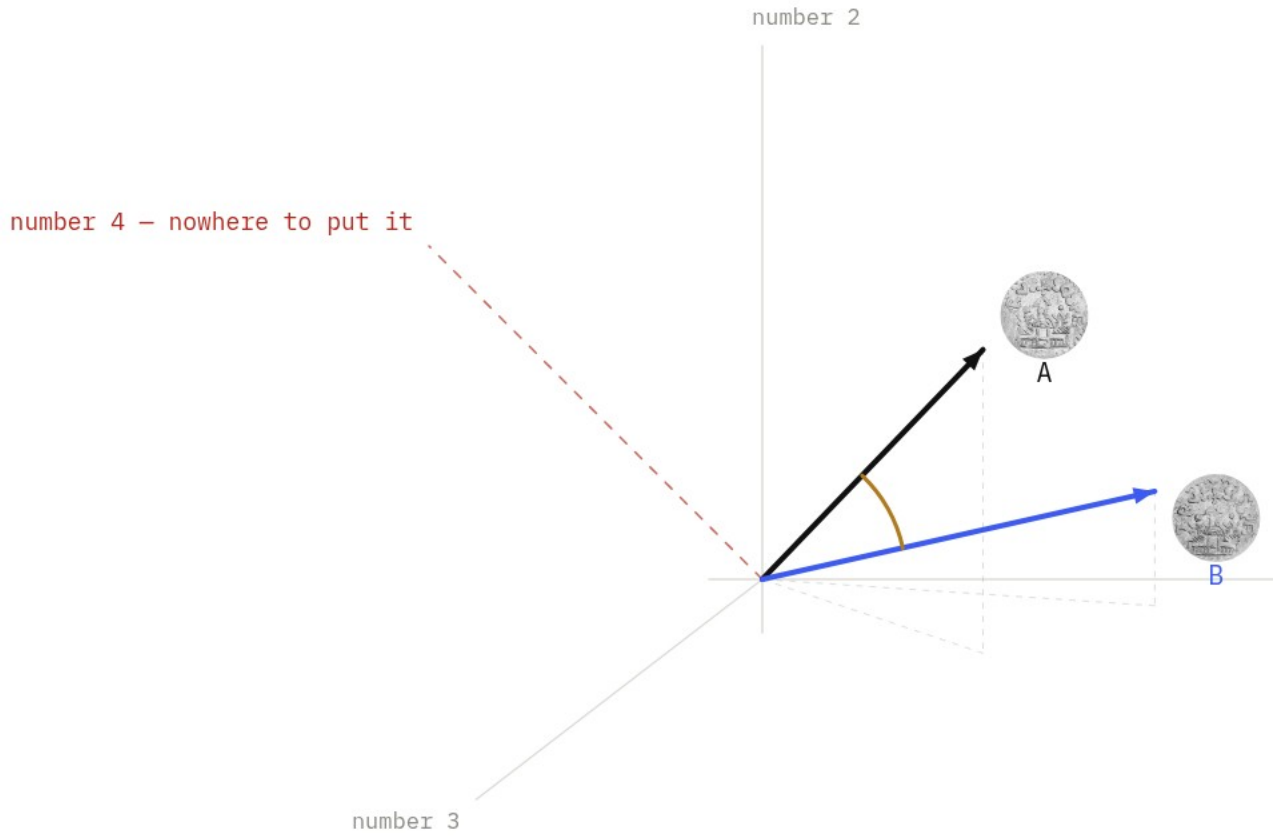
Where the idea came from

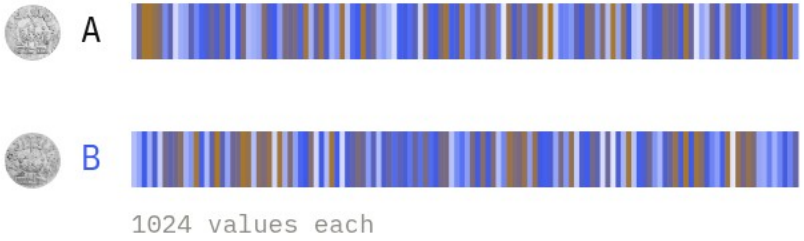
You hand it a photograph and get back similar photographs — and you never told it what similar means.

Nobody wrote the rule “a cat has ears”. The model learned to turn an image into **numbers**, arranged so that images that look alike end up with numbers that are close together.

If that works for cats and cathedrals — why not for coins?

The same measurement, however many numbers





cos **0.91**

the angle between the two arrows — the same arithmetic, whatever the count

So: 1024 of them.

Undrawable, and no harder to compute than two. **That** is why a thousand.

A model that was never told what anything is

Trained on nothing we have

millions of ordinary photographs, no labels at all

By filling in gaps

hide part of the picture, predict what was there

We borrow the sense, not the answer

it never saw a coin called a coin

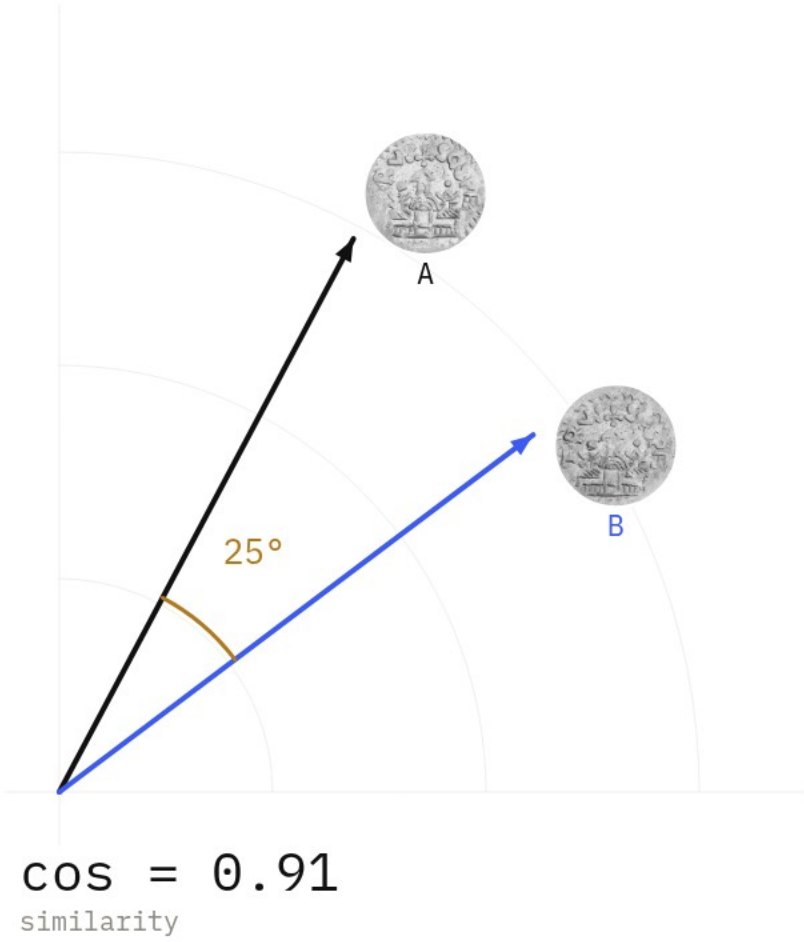
Several were tried. This one gave the most consistent neighbours — **chosen by measurement, not by the paper it came with**.

How a machine sees an object

196 patches



1024
numbers



cos = 0.91
similarity

Similarity is the angle between the arrows.

0.91 — and that is every measurement the machine will ever make about these two.

Live: who ends up next to whom

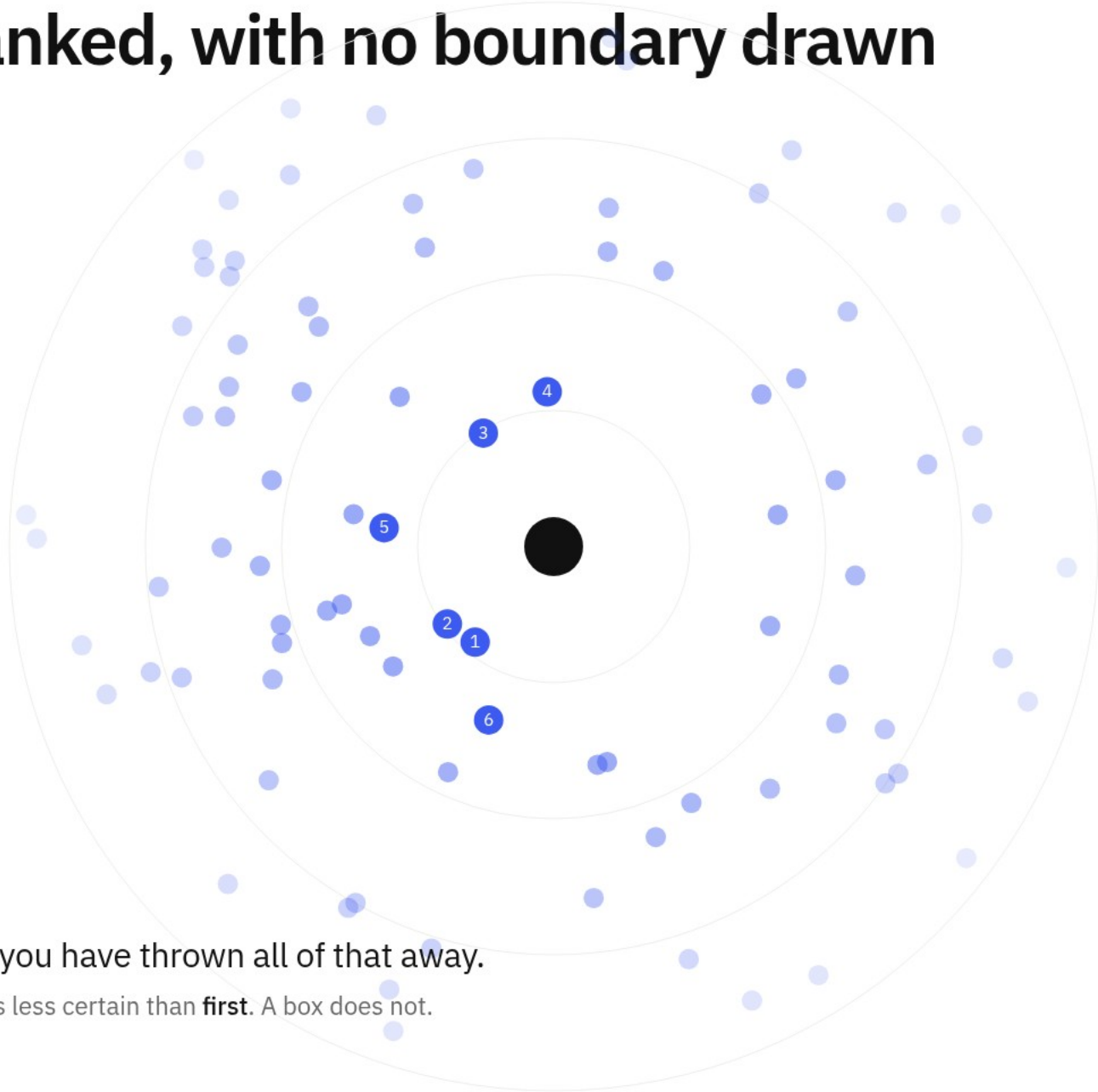
LIVE · IN THE APPLICATION

Open “Prohlížet” ↗

http://localhost:5173/app/datasets/2/explore

1. Left menu → **Prohlížet**. The whole find, placed by vector alone
2. Let the room pick a coin. Click it
3. Its neighbours and links appear — **would you have grouped those?**
4. Do it again with a badly worn one

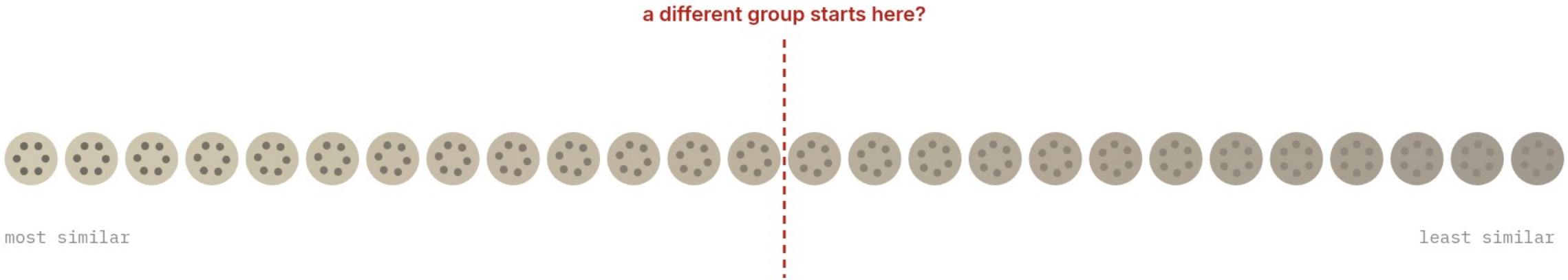
Neighbours ranked, with no boundary drawn



Draw one boundary and you have thrown all of that away.

The ordering knew that **sixth** was less certain than **first**. A box does not.

Where is the line?



Wherever you put it, you have just separated two coins that look the same.

CUT

Two families of clustering

Divide everything

every object is placed — including the ones that belong nowhere

Find the dense places

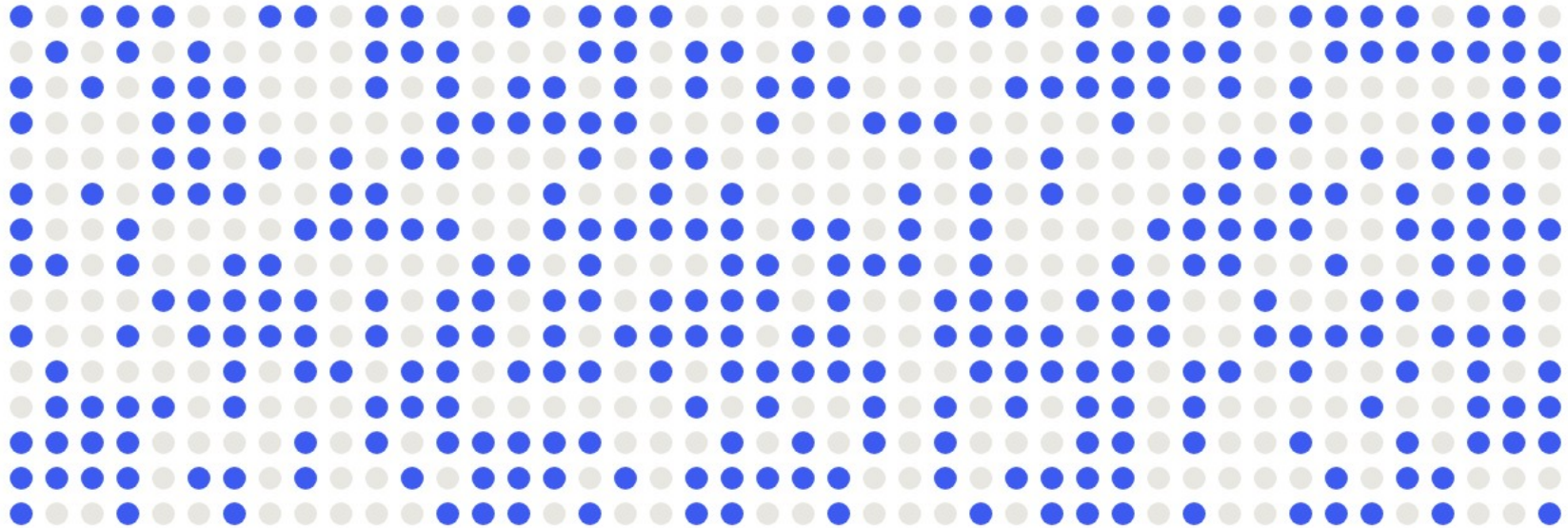
dense regions become groups. The rest is left **unassigned**.

HALF THE FIND CAME BACK UNASSIGNED

Everybody read that as a failure. Including me.

It is the only honest part of the output. A method that places every object gives you a cleaner-looking result that is **more wrong** — and gives you no way of telling which objects it made up.

317 coins left unassigned



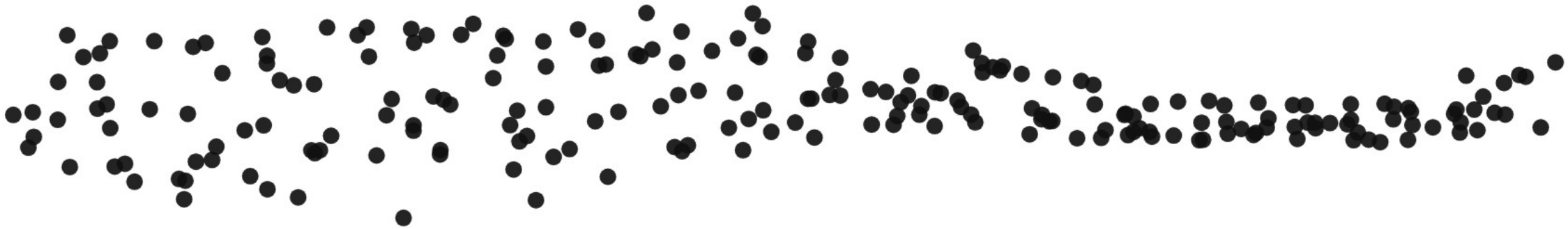
317

“I don’t know”

And declined to place these.

Not an error. **The only honest part of the output.**

Drag the threshold and watch the group count



1
groups

everything is its own group

everything is one group

Two hundred and twenty objects, no gap anywhere in them.

Drag until the grouping looks right. **Shout when it does.**

Reading the map, carefully

LIVE · IN THE APPLICATION

Open the map again ↗

http://localhost:5173/app/datasets/2/explore

1. Back to **Prohlížet** — the same map, now that we know what made it
2. Read the line the application prints under the title, out loud
3. Zoom in and out. **Nothing about the picture changes what is true**
4. Point at two islands side by side and ask whether they are related

Three things a UMAP picture does not mean

Distance between islands means nothing

neighbouring islands are not related

The size of an island means nothing

spread is an artefact of the drawing

Run it again and it changes

same data, different picture

It is a way of looking. It is not a result.

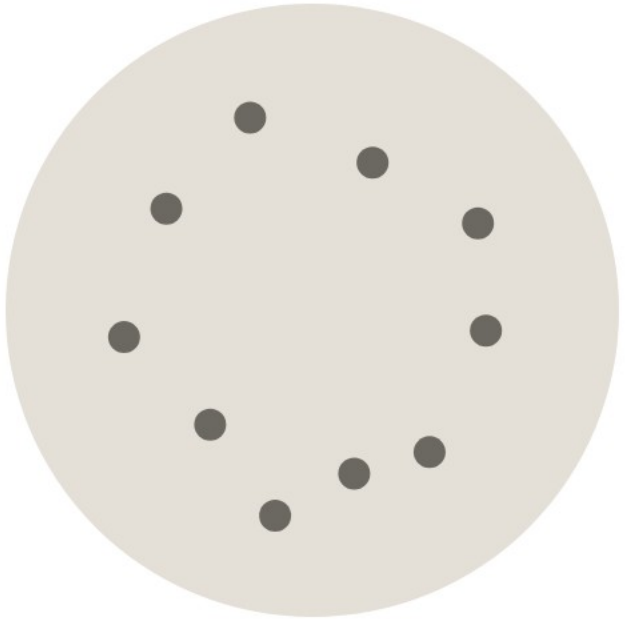
Let’s go to Mentimeter



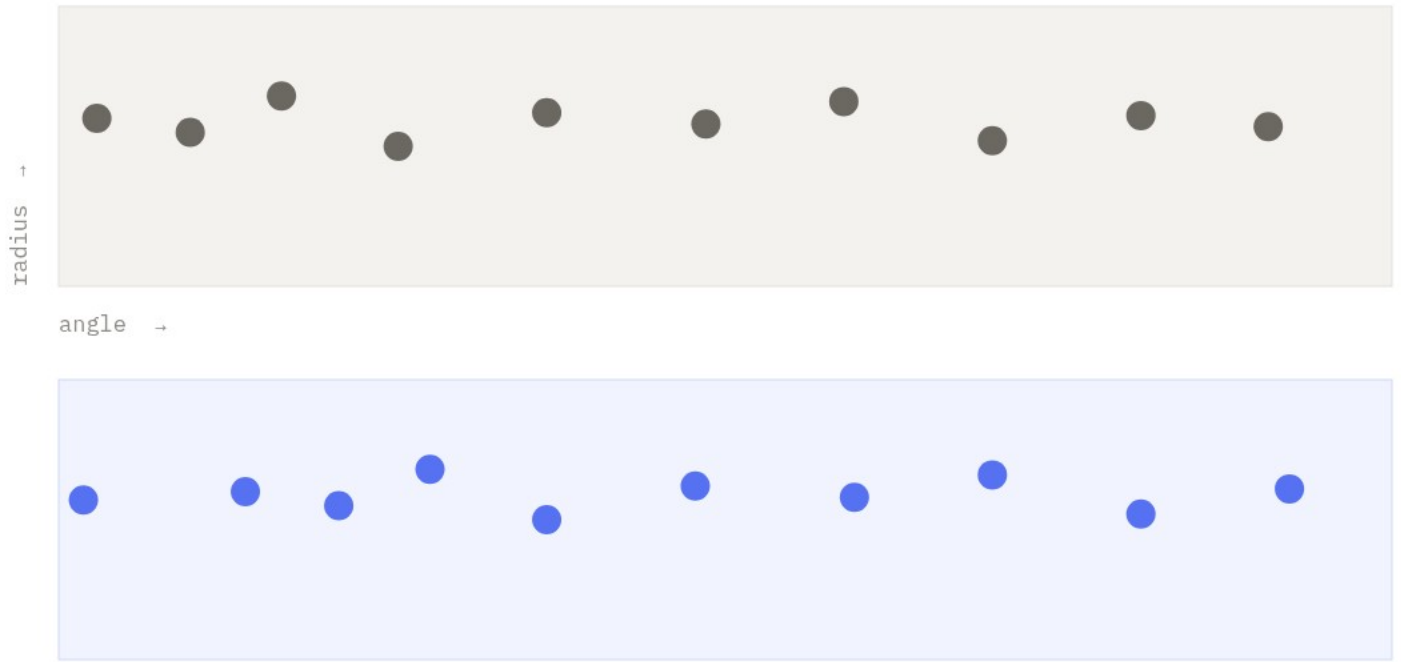
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Attempt one: unroll the circle



the coin



turning the coin = sliding the strip

It never beat the plain embedding.

Attempt two, and then the cheat

Embed it eight times

once per rotation, average the result — it works, and it costs eight times as much for a problem that need not exist at all

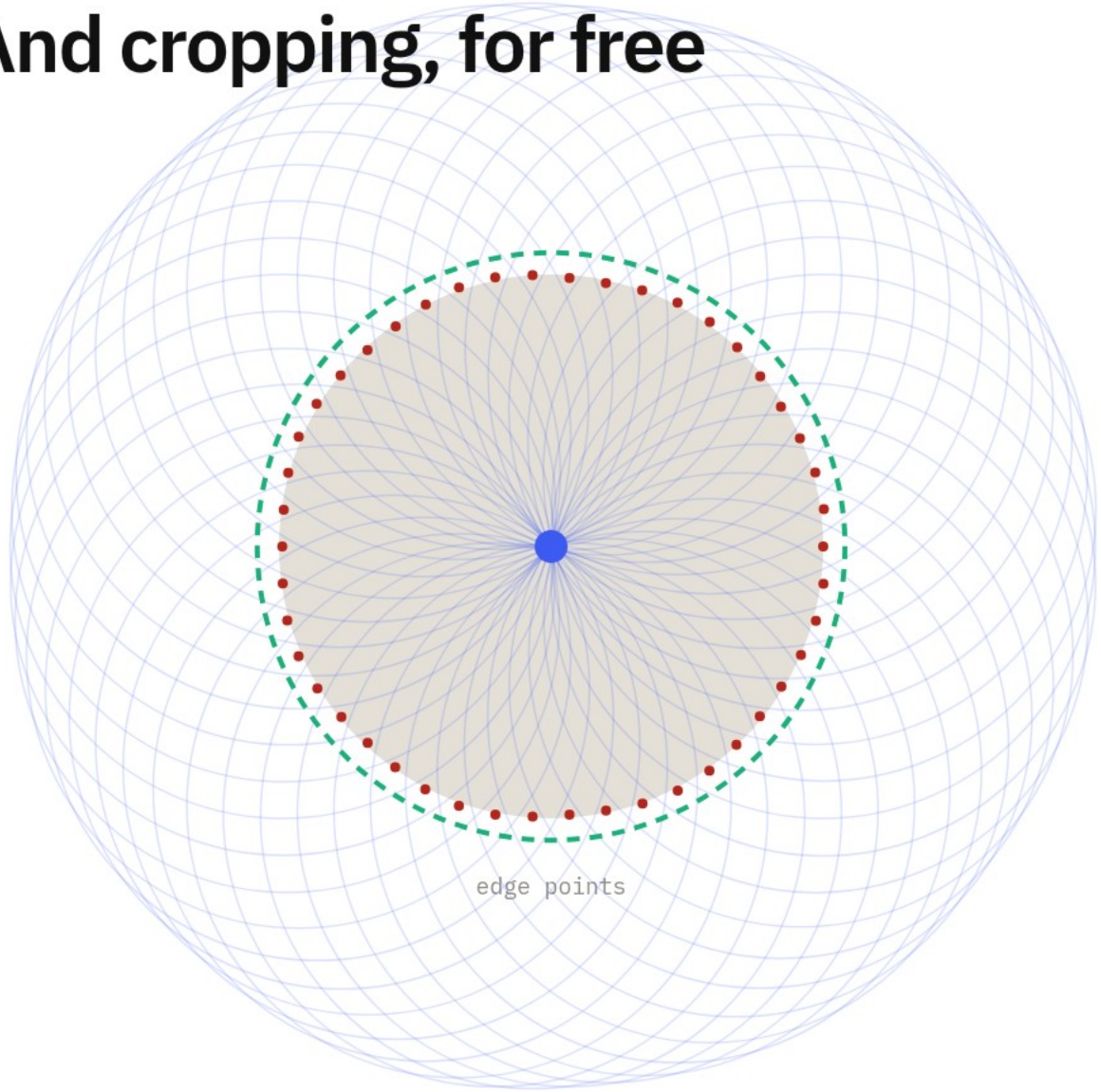
Or cheat

the dataset was already oriented, so I used that and moved on

A deferred problem you know about is a **decision**. A problem you do not know about is a **hole**.

The real fix is a small model that straightens a coin once, on the way in.

And cropping, for free



every edge point votes for where the centre could be

they agree here

crop here, with a margin

Live: and a person finishes it

LIVE · IN THE APPLICATION

Open “Mince” ↗

http://localhost:5173/app/datasets/2/coins

1. Left menu → **Mince**, open any coin
2. Show the crop — the circle from the vote, plus a margin
3. Adjust it by hand, then **Potvrdit**
4. No damage detector yet — **deliberately**, for now

What it can do	Find similar coins, with no training and no rules written by anyone.
What it is really asking	“Do these two photographs look alike?” — photographs, not coins.
Where it breaks	There is no line in a gradient. And it attends to damage, lighting and rotation, because nothing told it not to.

ACT VI

The image on a coin is not colour

It is shape. So let us have the shape.

I II III IV V **VI** VII VIII IX X

1:40 — 14 minutes

Let’s go to Mentimeter



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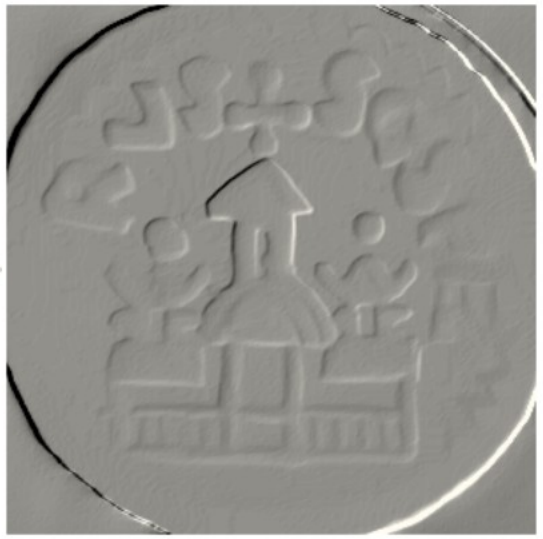
Guessing shape from one photograph



the photograph



estimated height, per pixel



lit from the side — the relief comes back

Look at the rim: the estimate gives up there.

It does that on every coin, in the same way — which turns out to be what matters.

We do not need it to be right. We need it to be wrong in the same way every time.

We are not measuring a coin. We are comparing two coins — and a poor representation compares perfectly well, as long as it is poor consistently.

A person judges a representation by **“does it look right”**. A comparison judges it by **“does it behave the same way every time”**. Those are not the same question.

Two channels, and neither one wins

From the photograph	From the depth
colour, texture, and everything that happened to the coin	the relief of the strike, with the colour of the metal gone

Neither is reliably better. Which one wins depends on the individual photograph — so both are kept.

And an open problem: when depth estimation fails, it fails **quietly** — a smooth, plausible, empty surface. Coins that failed the same way then resemble **each other**, for reasons that have nothing to do with coins.

What it can do	Recover the shape of the strike from an ordinary photograph, with no scanner.
What it is really asking	“What shape would explain this shading?” — a guess, and it knows nothing about coins.
Where it breaks	Quietly. A failed depth map looks like a fine depth map.

ACT VII

A gradient is not a group

Everything so far works. None of it answers the question.

I II III IV V VI VII VIII IX X

1:54 — 12 minutes

Where we had got to

We can find similar

and the neighbours are good

We have two channels

photograph and depth

We can draw a map

and read it, carefully

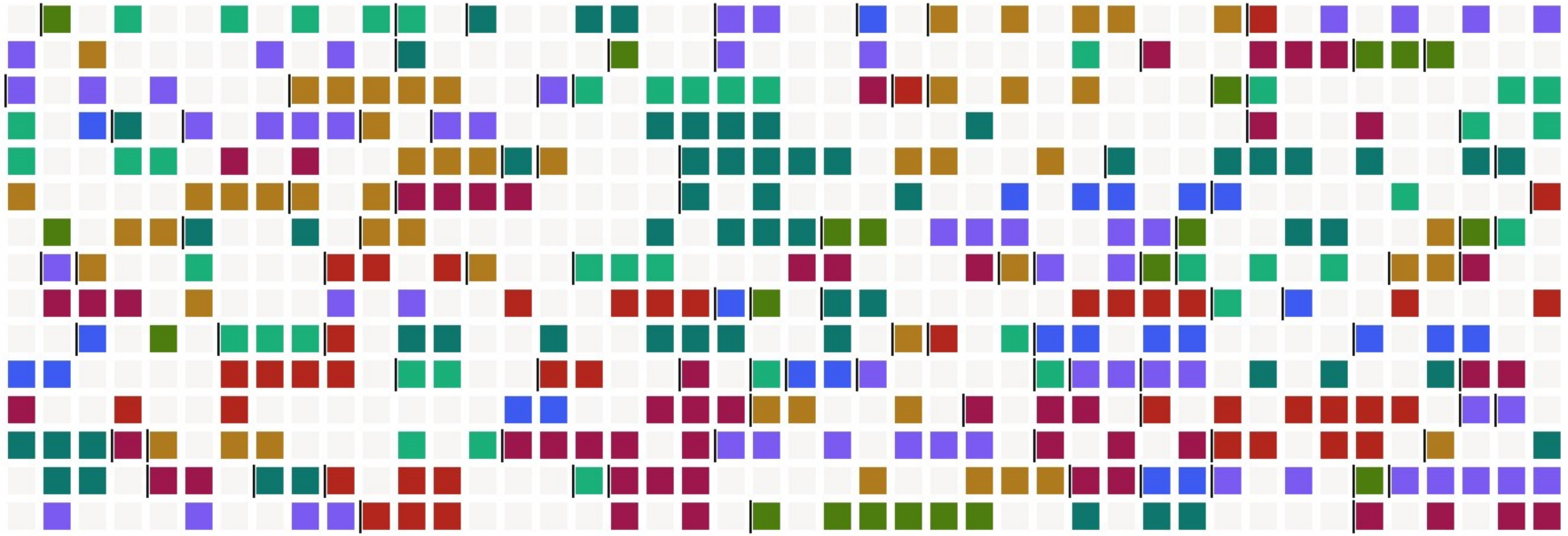
And we still have **no groups** — and no dies.

We asked a numismatist

> **“The groups resemble each other. And inside each one there are several different dies.”**

242 groups for 660 coins — and more dies than groups. What the embeddings divide on is **overall appearance**: wear, centring, lighting. Reproducible, and numismatically meaningless.

242 groups, and every one of them pure



660
coins

242
groups

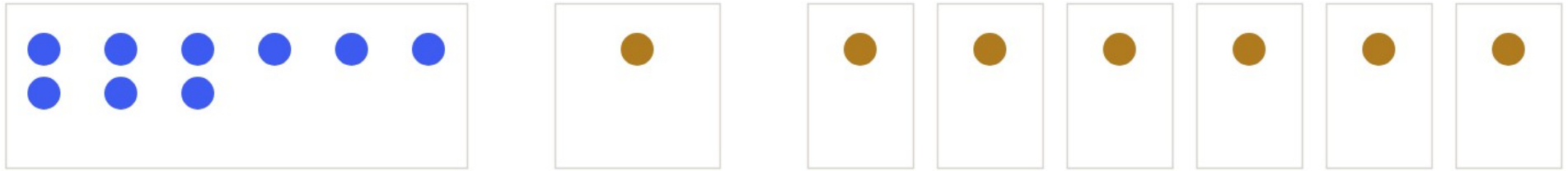
48%
unassigned

8
types the specialist used

Now colour every coin by the specialist’s type.

Not one group mixes two colours. Every group is pure — and there are 242 of them for 8 types.

Why a purity score rewards splitting



average purity: 100%

number of useful classifications: zero

Split finely enough and the score goes to 100% by construction.

242 groups. Half the coins dropped. A perfect number, measuring nothing.

Let’s go to Mentimeter



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Built for exactly this, and it did not work

Face recognition. Siamese networks. Architectures built for precisely this question.

A face has landmarks that are **always there**. Two eyes, a nose, a mouth, in an arrangement that survives lighting and age.

A worn coin is faded

part of the device is simply gone — and it is gone somewhere different on every single coin

What it can do	Group coins by how they look, reproducibly, with no training.
What it is really asking	“Which of these photographs resemble each other?” It was never asking about dies.
Where it breaks	It does not break. It works, and contributes **nothing** to the question we were paid to answer.

ACT VIII

Does a coin have a fingerprint?

The question that finally had the right shape.

I II III IV V VI VII **VIII** IX X

2:06 — 24 minutes

Everyone knows this from television

Part of a fingerprint is enough to identify a person — and a print is only ridges and the places they split.

It is the **arrangement** that identifies, not the ridges.

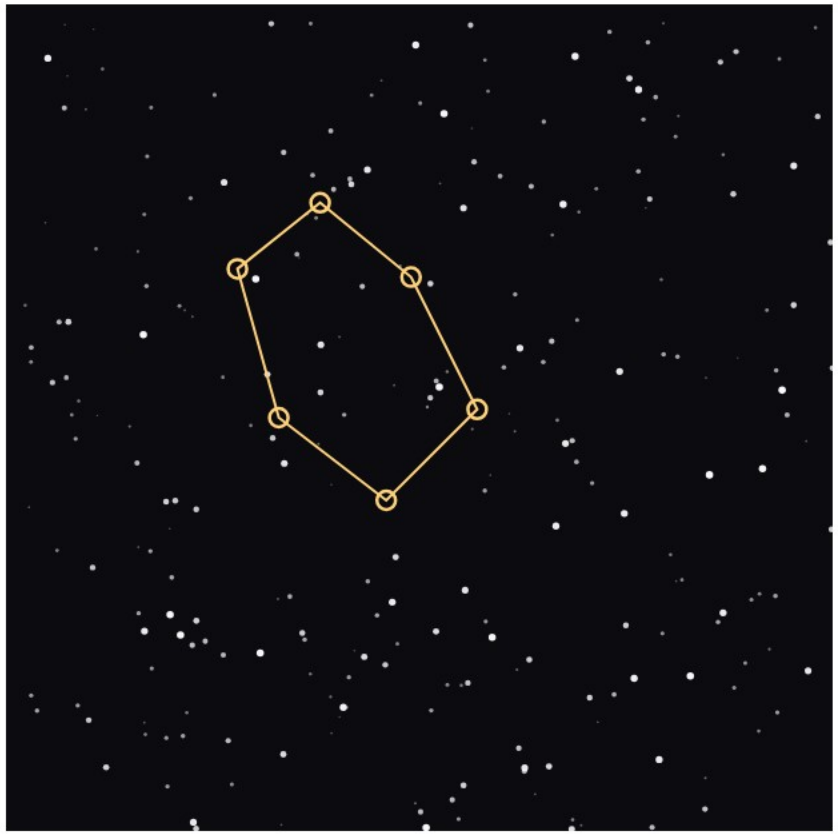
So I tried fingerprint algorithms

they look for ridge flow and where it forks. A coin has no ridges. Nothing.

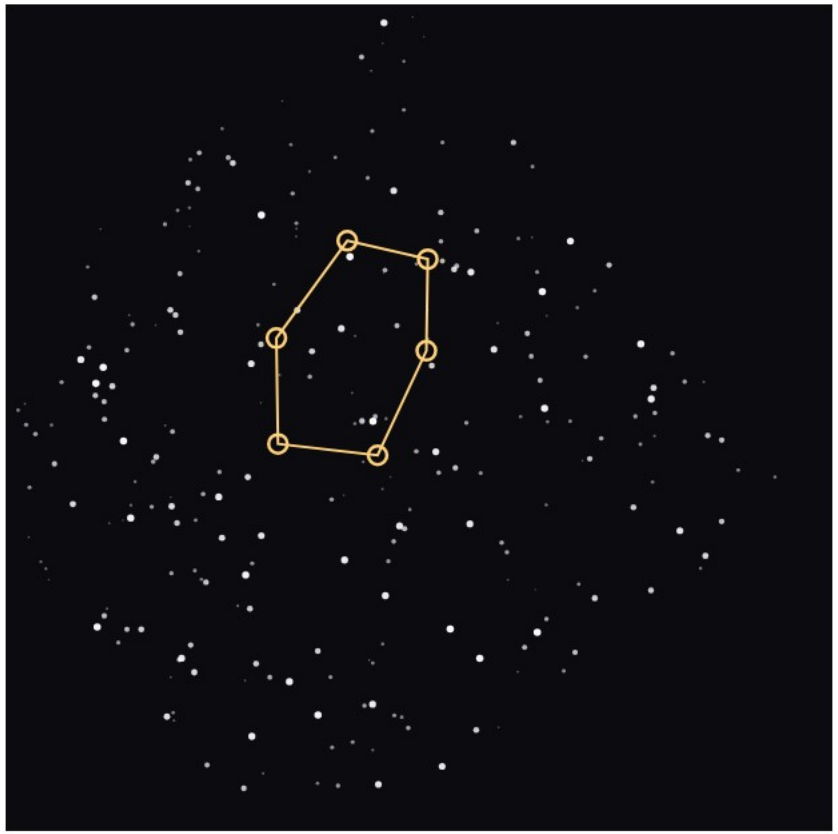
Then matching the shapes themselves

a worn letter is a different shape on every coin. Nothing.

Astrometry: matching by arrangement



the photograph



the catalogue

Look that arrangement up.

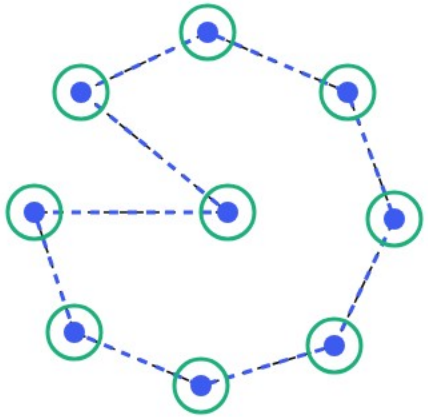
It is the same shape, turned and scaled. **Now you know where you were pointing.**

Spring came, and I started paying more attention to what is over our heads.

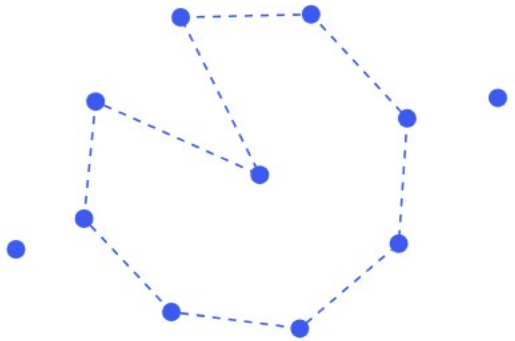
An astronomer has a patch of sky and no idea which patch. There are billions of stars in the catalogue.

They do not compare stars. They compare **how the stars sit relative to one another** — ratios that survive turning, moving and scaling.

What if every struck island is a star?



coin A



coin B

nine marks, the same arrangement

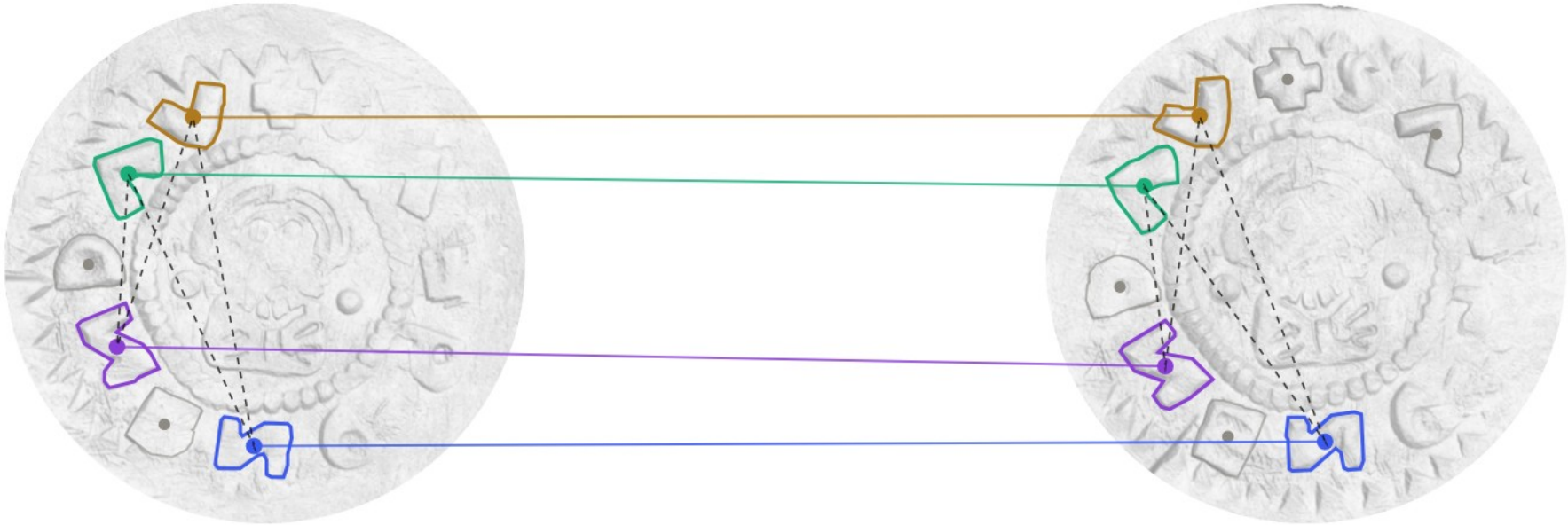
And how do you find the islands?

You ask for them in words.

A segmentation model that takes a text prompt — *“letters, symbols, ornaments”* — and returns the regions that match it.

No training	**No annotation**	**Nothing tuned to this box**
constraint 2, satisfied	constraint 2, again	constraint 1, satisfied

A real match, and how it was decided



S 8371 · 6 regions found

S 8372 · 8 regions found

4 in agreement · turned 10.5°, scaled 1.043

Four of them sit in the same arrangement.

The coin is turned ten and a half degrees. Four marks agree anyway — and the two that do not are simply left out.

Why it survives wear and off-centre strikes

Wear blurs outlines, does not move marks	**Struck off-centre** three surviving marks still make the claim	**Struck at an angle** the transform absorbs it

217 470 every possible pair in the find	**12 s** to check all of them, on a laptop	**25** pairs that matched

And it worked

217,470	12 s	25	25 / 25
pairs	for all of them	matches	inside the same expert type

By chance, that last one has a probability of about 10^{-39}.

The **cheapest** method of the afternoon is the only one that answers the question we were asked.

So why not be less strict?

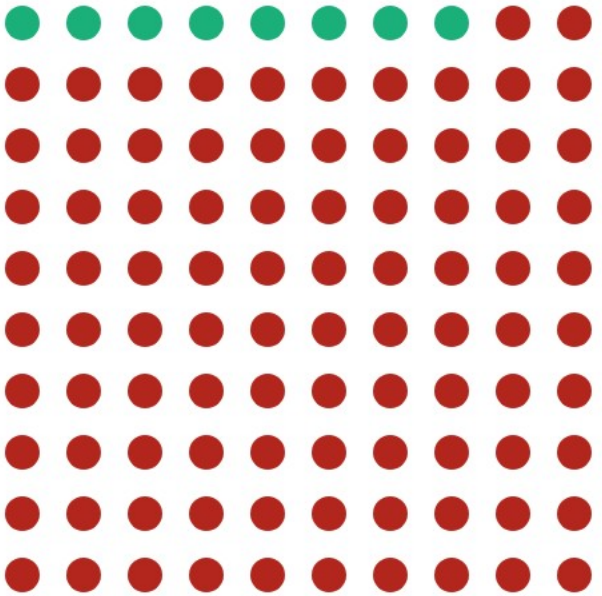
As it is set now



25 pairs reported out of 217,470

every one inside the same expert type

Loosened until it finds more



for every 100 pairs it now reports…

…8 are right, and 92 are not

And “these two look alike” is right 88 % of the time, for free.

Live: how it got there

LIVE · IN THE APPLICATION

Open “Razidla” ↗

http://localhost:5173/app/datasets/2/die-check

1. Left menu → **Razidla** → **Porovnat konstelace prvků**
2. Open one match
3. Matched regions coloured and numbered, unmatched grey
4. The constellation, and the table of pairs underneath

What it can do	Show that one tool touched two objects — the strongest claim we can make from photographs.
What it is really asking	“Do a handful of marks sit in the same arrangement?” Geometry, not resemblance.
Where it breaks	When there are no marks to find: a poor photograph, heavy wear, a plain surface.

ACT IX

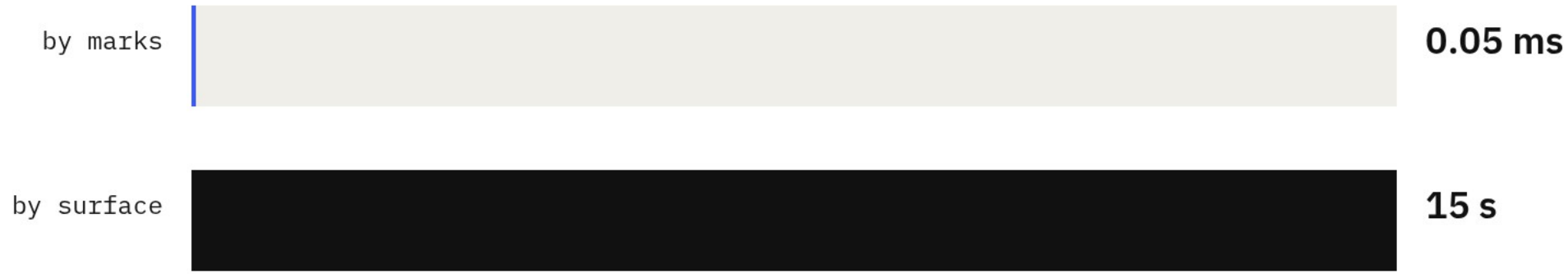
The last witness

The strongest thing we can say, and the most expensive.

I II III IV V VI VII VIII **IX** X

2:30 — 16 minutes

Cost per pair: 0.05 ms against 15 seconds



All 217 470 pairs, both ways.

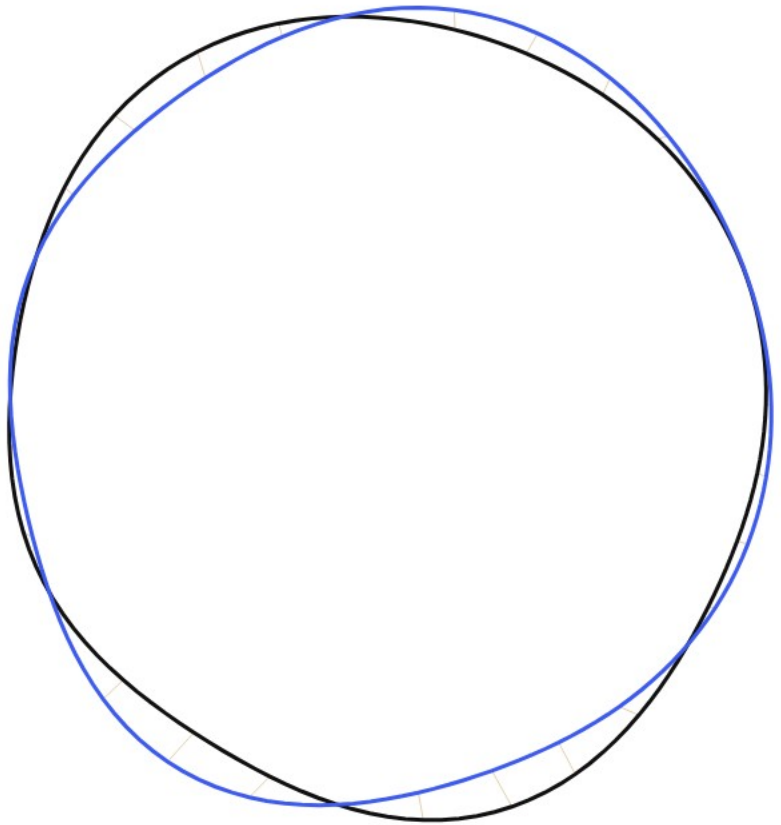
Twelve seconds against **a hundred and three years**. That is what decides the order.

It confirms. It does not search.

Two surfaces, brought into the same frame, and the distance between them measured point by point.

Twenty seconds a pair. It runs on candidates the cheaper stages have already argued for — **never as a way of finding them**.

Bringing two surfaces into register



surface A

surface B

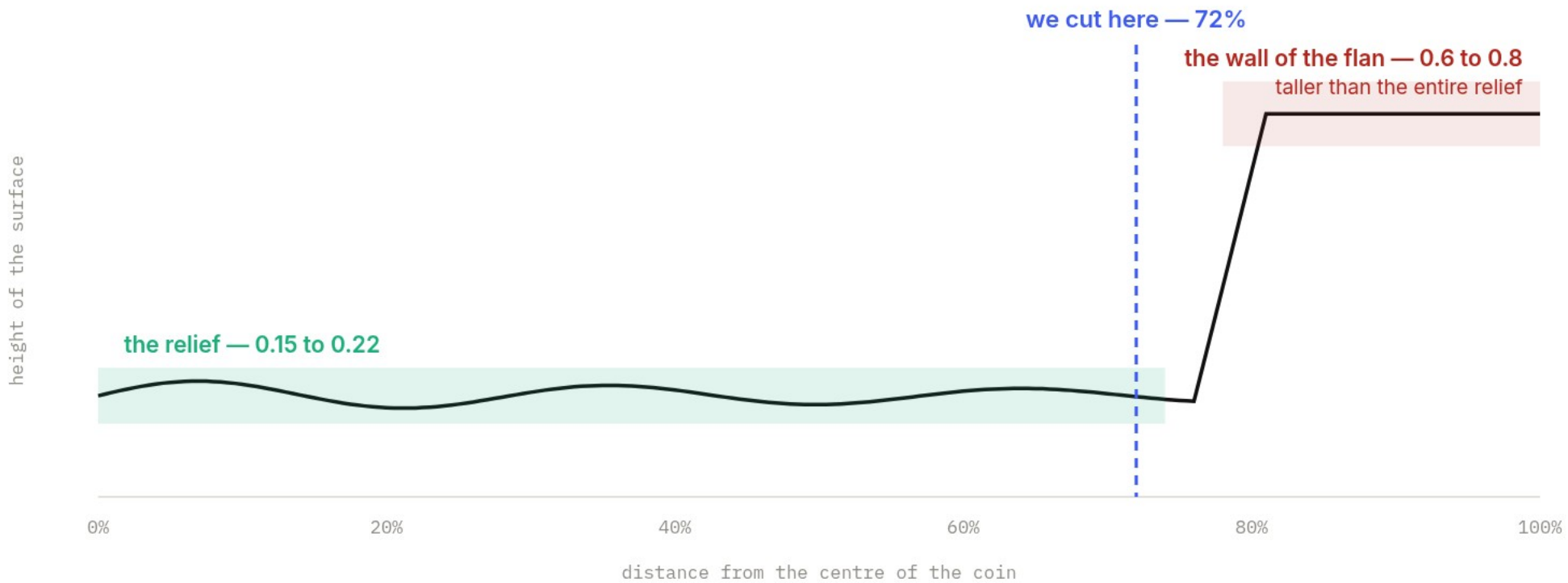
26

iterations

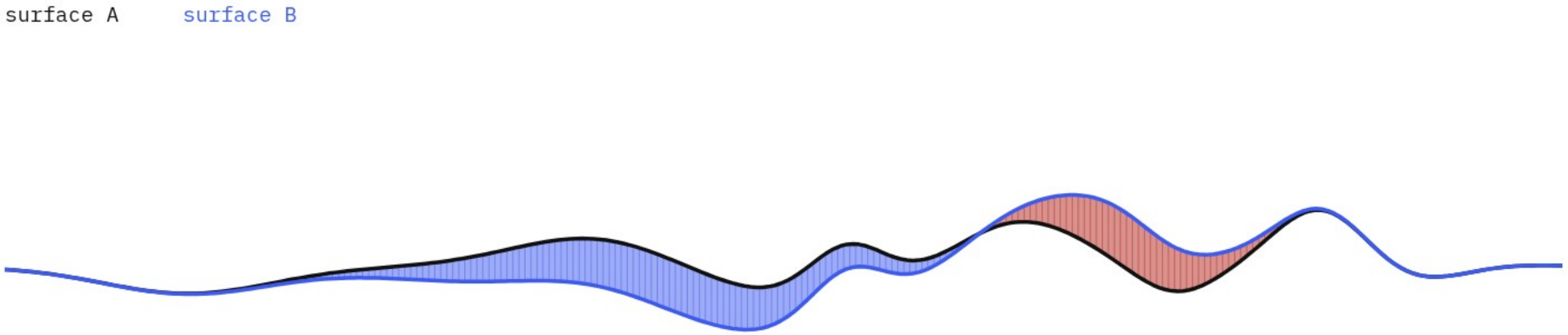
That is the whole algorithm. It is from 1992.

One catch: it finds the nearest fit to **where you started it**. Start it badly and it converges, confidently, to nonsense.

The dullest part decided the result



Signed distance: which surface stands higher



That sign is what the colour map is made of.

Worn in the middle, one extra lump on the right. An unsigned distance would make those look the same.

Live: where it agrees and where it does not

LIVE · IN THE APPLICATION

Open “Razidla” ↗

http://localhost:5173/app/datasets/2/die-check

1. Left menu → **Razidla** → **Porovnat dva povrchy**
2. Pick two scanned coins and compare
3. **Sedí / nesedí**, then drag **Tolerance**
4. Click the coin for **Řez mincí** — the section through both
5. Say it plainly: the colour scale is relative to **this pair alone**

We tried to derive a threshold

Distance alone	Shape agreement alone	Both together
6 of 31 others inside the range of the 5 known matches	11 of 31	4 of 31 — and no better line exists

Five positives is very few. “No match” is not proof of a different die. And one section samples very little surface.

So there is **no threshold** — and that is a result, not a gap in the work.

You cannot trust a result because it looks reasonable. You can only trust a contradiction that cannot be true.

For a month those numbers looked sensible. The library was overwriting its own input — only above fifty thousand elements, so every small test passed.

What caught it was one printed value that could not follow from the values above it.

What it can do	Compare the objects themselves, and show a person exactly where they differ.
What it is really asking	“How far apart are these two surfaces, once aligned?” — after a rigid alignment that cannot model a bent coin.
Where it breaks	On preparation, on library versions, and on anybody reading the colour instead of the number.

ACT X

The wrong question

Three hours to arrive at what should have been asked first.

I II III IV V VI VII VIII IX X

2:46 — 14 minutes

Let’s go to Mentimeter



MENTI.COM

8419 1952

What we actually built

Rank by similarity	Draw a map	Assert 25 facts
useful, and it decides nothing	readable, if you know what it does not mean	out of 217,470 pairs — one tool touched both

We **cannot** classify coins automatically. We never could.

Classification is something people do. And people do not agree.

Two specialists, one tray, two classifications — both defensible, neither provable. There is no answer key.

So an algorithm that classifies automatically does not remove the disagreement. It **freezes one person's version of it** and gives it the appearance of objectivity.

Let’s go to Mentimeter

MENTI.COM

8419 1952

What the classification was for



Dating

when was it in the ground

Relationships

what was struck together, and in what order

Movement

trade, an army, a gift — how it travelled

a shared die answers this directly

one place, one moment — with no classification on the way

We asked how to automate the intermediate step. We should have asked what the intermediate step was for.

Seven questions to ask

1. **What question does this answer — and is it mine?**
   act I · the one I did not ask
2. **What exactly does it see when I give it my material?**
   act II · everything except the point
3. **What does it do when the object is damaged?**
   act IV · it found the corrosion
4. **Show me a case where it fails.**
   act VII · it worked, and contributed nothing
5. **What does it cost on a thousand? On a hundred thousand?**
   act IV · cost decides before accuracy
6. **Whose judgement is built into it?**
   act X · one person’s version, frozen
7. **When it says “the same” — what exactly is it claiming?**
   act VIII · geometry, not resemblance

It was not a better model. It was finally asking, after three hours, the question we should have asked first.

Your box is not coins. Everything here transfers.